

Embedded Linux and Real-Time Operating System Development for Internet of Things Applications

Youssef Khaled Omar Mahmoud

Department of Electrical, Electronic and Communication Engineering

Faculty of Engineering, Cairo University

Giza, Egypt

youefkh05@gmail.com

ORCID: 0009-0005-5807-4381

Abstract—This paper presents the work completed during the IoT Summer Research Internship 2026 at the Nanoelectronics Integrated Systems Center (NISC), Nile University. The internship focused on embedded systems, Internet of Things (IoT), and real-time operating systems through a series of hands-on tasks. The developed solutions demonstrated communication between embedded devices and cloud services while providing practical experience in embedded software development, cloud technologies, wireless communication, and system integration.

Index Terms—Internet of Things (IoT), Embedded Systems, Raspberry Pi, Embedded Linux, TI-RTOS, CC1352P1 LaunchPad, MQTT, HTTP, InfluxDB, Grafana, UART, Bluetooth, Zigbee.

I. INTRODUCTION

The Nile University Summer Research Internship 2026 provided an opportunity to gain practical experience in embedded systems, Internet of Things (IoT), and real-time software development through a series of hands-on engineering tasks. The internship emphasized applying theoretical knowledge to real-world embedded applications while introducing modern software and communication technologies widely used in industry.

Throughout the internship, three major technical areas were explored. The first focused on Embedded Linux development using the Raspberry Pi, where Python was employed to interface with hardware peripherals and implement different GPIO control techniques. The second addressed the design of an end-to-end IoT system by transmitting sensor measurements from edge devices to cloud services using both HTTP and MQTT communication protocols. The collected data were stored in a time-series database and visualized through a real-time monitoring dashboard. Finally, the internship introduced real-time embedded software development using TI-RTOS on the Texas Instruments CC1352P1 LaunchPad, where multiple concurrent tasks including LED control, Bluetooth communication, Zigbee communication, and UART-based communication with the Raspberry Pi were implemented.

In addition to software implementation, the internship emphasized understanding the design trade-offs between different communication protocols and software architectures. The progression from basic hardware interfacing to cloud connectivity

and finally to multitasking embedded systems provided a comprehensive view of modern IoT system development.

This report summarizes the assigned tasks, describes the methodologies followed during implementation, presents the achieved results, and discusses the technical knowledge and practical skills acquired throughout the internship.

II. ASSIGNED TASKS

During the internship, the following technical tasks were assigned and successfully completed: [1]

- Raspberry Pi development using Embedded Linux.
- Python programming for hardware interfacing.
- GPIO programming and sensor interfacing.
- Cloud communication using HTTP.
- Cloud communication using MQTT.
- Data storage using InfluxDB.
- Data visualization using Grafana.
- Real-Time Operating System (RTOS) programming.
- Bluetooth communication.
- Zigbee communication.
- UART-over-USB communication.
- Integration between the RTOS microcontroller and the Raspberry Pi.

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